

TAWATINAW AGCM, AB, CanadaWMO: **715490**Lat: **54.300N**Lon: **113.520W**Elev: **611**StdP: **94.20**Time Zone: **-7.00 (NAM)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-31.0	-27.9	-34.6	0.2	-30.7	-31.5	0.2	-27.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	27.1	16.9	25.6	16.2	24.1	15.5	18.9	24.4	17.7	23.3	16.7	22.1	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.9	22.1	15.5	11.9	20.4	14.6	11.1	19.2	56.4	24.5	52.4	23.4	49.3	22.2	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-34.5	30.4	2.4	0.7	-36.2	30.9	-37.6	31.3	-38.9	31.7	-40.6	32.1
			-34.6	21.5	2.3	1.6	-36.3	22.7	-37.6	23.7	-38.9	24.6	-40.6	25.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.7	-11.0	-11.2	-4.9	3.4	11.1	14.8	16.7	15.6	10.9	3.9	-4.9	-12.7	(6)
	DBStd	12.24	8.68	8.09	7.49	5.06	4.65	3.13	2.60	3.18	4.49	4.78	6.80	7.50	(7)
	HDD10.0	3348	649	593	461	205	44	4	0	2	42	195	448	705	(8)
	HDD18.3	5730	908	827	719	449	228	112	63	93	224	446	698	963	(9)
	CDD10.0	692	0	0	0	6	77	147	209	177	69	7	0	0	(10)
	CDD18.3	33	0	0	0	0	2	5	14	10	2	0	0	0	(11)
	CDH23.3	482	0	0	0	2	60	78	171	122	49	1	0	0	(12)
	CDH26.7	57	0	0	0	0	9	7	21	13	7	0	0	0	(13)
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
(15) Precipitation	PrecAvg	431	22	11	17	25	49	82	82	56	35	19	20	18	(15)
	PrecMax	538	93	23	38	49	79	134	161	118	79	34	70	33	(16)
	PrecMin	321	0	0	3	1	7	32	38	22	4	2	3	1	(17)
	PrecStd	71	22	7	10	13	19	29	31	22	19	10	17	10	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	6.2	11.3	22.5	28.0	27.5	28.7	28.2	27.6	21.3	11.4	4.6	(19)
		MCWB	2.7	2.8	5.7	10.5	13.1	15.9	19.8	17.7	16.0	11.9	6.6	1.6	(20)
	2%	DB	4.6	3.7	7.9	17.4	24.9	25.5	26.9	26.4	24.5	16.5	7.5	2.2	(21)
		MCWB	1.8	1.1	3.6	7.9	12.1	15.1	18.3	17.7	14.9	9.2	4.3	-0.4	(22)
	5%	DB	3.1	1.9	5.7	14.3	22.3	23.7	25.2	24.6	22.1	13.6	5.3	0.1	(23)
		MCWB	0.7	-0.3	2.4	6.6	11.1	14.3	17.3	16.8	14.0	7.7	2.6	-1.7	(24)
	10%	DB	1.2	0.2	3.8	11.7	20.2	21.9	23.6	22.9	19.1	11.3	3.0	-2.4	(25)
		MCWB	-0.6	-1.5	1.0	5.2	10.1	13.8	16.5	16.1	12.4	6.4	1.2	-3.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	3.0	6.2	10.9	14.3	18.1	21.2	20.3	17.2	12.3	7.1	2.0	(27)
		MCDB	5.6	5.6	10.1	22.1	22.7	24.1	27.1	25.7	25.3	19.9	10.7	4.3	(28)
	2%	WB	2.1	1.3	4.0	8.4	13.2	16.7	19.4	18.7	15.7	10.0	4.6	-0.1	(29)
		MCDB	4.3	3.3	7.3	16.4	22.0	22.5	25.0	24.0	22.6	15.0	7.3	1.9	(30)
	5%	WB	0.9	-0.1	2.5	6.8	12.1	15.6	18.2	17.6	14.2	8.3	2.8	-1.9	(31)
		MCDB	2.9	1.7	5.4	13.3	20.8	21.3	23.8	22.8	21.0	12.9	4.8	-0.1	(32)
	10%	WB	-0.6	-1.6	1.2	5.6	11.0	14.7	17.1	16.5	13.0	6.9	1.2	-3.7	(33)
		MCDB	1.3	0.2	3.6	10.8	18.3	20.2	22.3	21.7	18.4	10.5	2.9	-2.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	9.1	8.9	10.2	12.4	11.1	11.4	11.7	11.8	9.3	7.1	7.5	(35)
		MCDBR	8.4	10.1	10.1	14.8	16.0	14.0	14.1	14.2	16.7	13.3	9.1	9.6	(36)
		MCWBR	6.3	8.1	6.6	7.7	6.7	5.9	7.0	6.6	8.0	7.4	6.4	8.2	(37)
	5% WB	MCDBR	8.7	9.8	9.8	13.9	14.5	12.1	13.0	12.9	15.6	12.7	8.5	9.2	(38)
		MCWBR	6.8	8.1	6.5	7.4	6.3	5.7	6.9	6.6	7.6	7.4	6.3	8.0	(39)
(40) Clear-Sky Solar Irradiance	taub		0.255	0.275	0.307	0.326	0.341	0.353	0.355	0.361	0.307	0.282	0.252	0.236	(40)
	taud		2.273	2.288	2.299	2.389	2.403	2.376	2.424	2.351	2.515	2.553	2.451	2.303	(41)
	Ebn at Noon		713	824	872	904	905	893	885	855	873	824	737	669	(42)
	Edh at Noon		56	79	99	104	109	113	106	107	80	61	47	44	(43)
(44) All-Sky Solar Radiation	RadAvg		0.86	1.88	3.40	4.65	5.68	5.94	6.01	4.93	3.40	1.91	0.89	0.62	(44)
	RadStd		0.10	0.14	0.26	0.39	0.47	0.40	0.32	0.35	0.35	0.25	0.10	0.05	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page